Auteur: Oubayy Ahale
Studierichting: Informatica
Oefeningenreeks 3E nr. 4

$$V(t) = 5000\left(1 - \frac{t}{40}\right)^2$$

$$V'(t) = 5000 \cdot \frac{d}{dt}\left[\left(1 - \frac{t}{40}\right)^2\right]$$

$$= 5000 \cdot 2\left(1 - \frac{t}{40}\right) \cdot \frac{d}{dt}\left(1 - \frac{t}{40}\right)$$

$$= 10000\left(1 - \frac{t}{40}\right) \cdot \left(-\frac{1}{40}\right)$$

$$= -250\left(1 - \frac{t}{40}\right)$$

Stel $Q(t) = 5000 - V(t)$

$$Q'(t) = -\left[-250\left(1 - \frac{t}{40}\right)\right]$$

$$= 250\left(1 - \frac{t}{40}\right)$$

4.1. Na 5 minuten:

$$Q'(5) = 250\left(1 - \frac{5}{40}\right) = 250\,(1 - 0.125) = 218.75 \text{ liter per minuut}$$

4.2. Na 10 minuten:

$$Q'(10) = 250\left(1 - \frac{10}{40}\right) = 250\,(1 - 0.25) = 187.5 \text{ liter per minuut}$$

4.3. Na 20 minuten:

$$Q'(20) = 250\left(1 - \frac{20}{40}\right) = 250\,(1 - 0.5) = 125 \text{ liter per minuut}$$

References